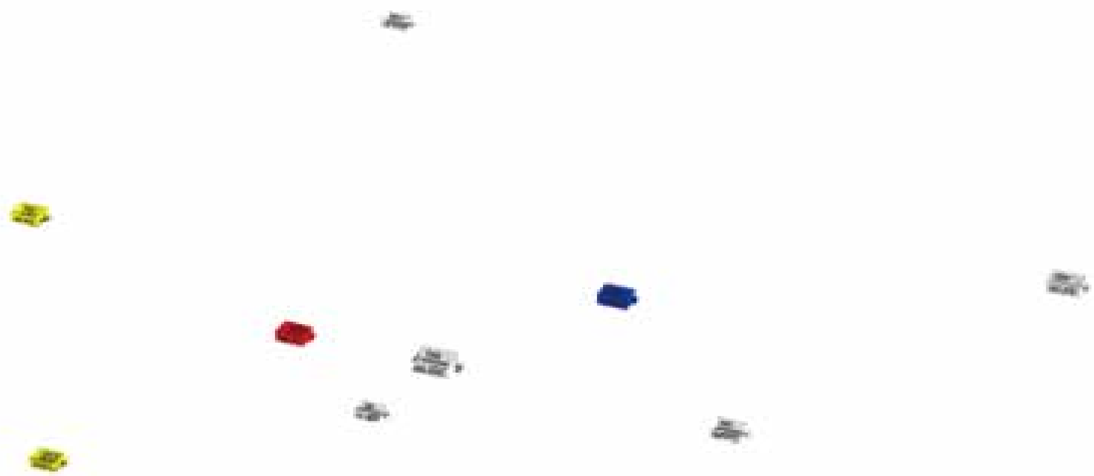
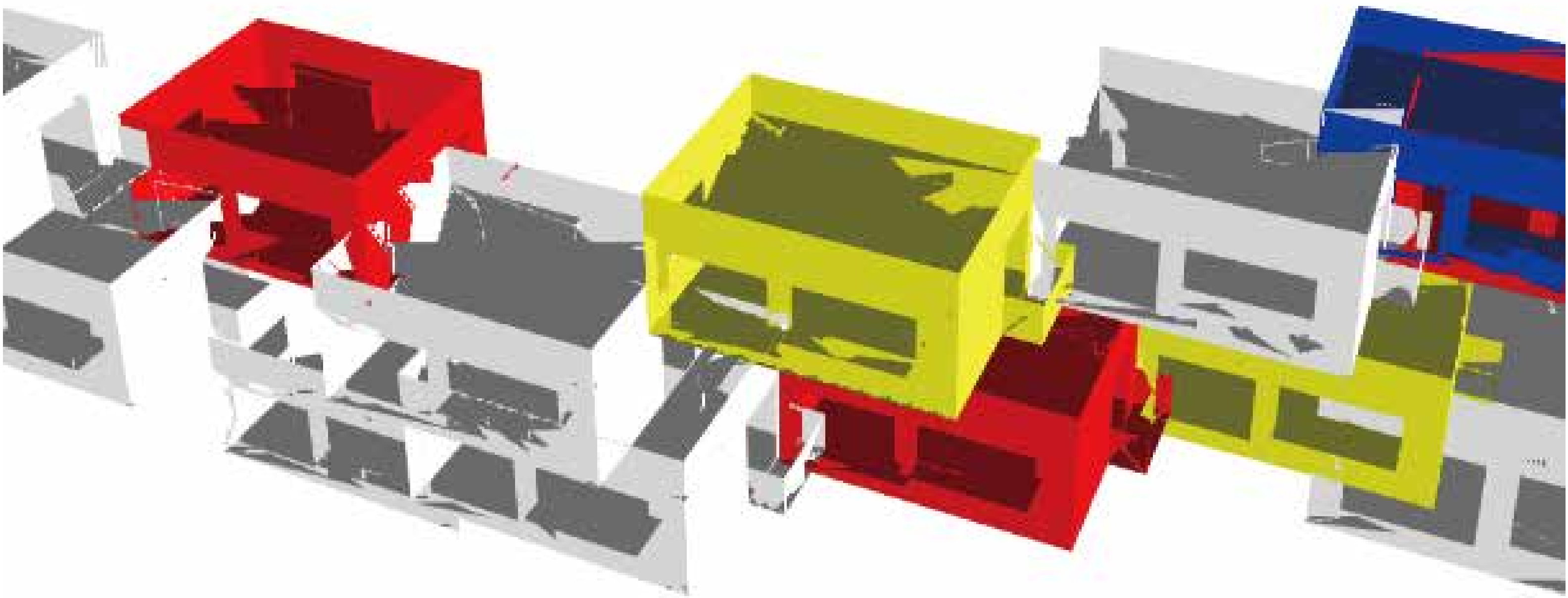
WHO DESIGNED THE SPACE?

Negotiate the algorithm to design the building

An architectural design simulator that recreates the design process within the context of the "collective practice" era.



Examples of work processes and references



Observation Report | Who designed the space?

HUMAN AUTHORSHIP 50%	AI AUTHORSHIP 50%	Data
		TRIM 2 UNDO 0 REMOVED 2

This is an architectual design with a equal negotiation.

Wow, an equal division of labor! Human-machine collaboration makes the work effortless!(*°▽°*)

Background research&Prototyping

1.1-See a fun topic

I saw an article about designer in Age of Collective Practice

"The question is no longer whether machines assist (designer) in design, but how much they participate in it; how architects should position themselves within this expanding ecology of human and non-human contributors."

—<The Machine in the Age of Collective Practice> 1

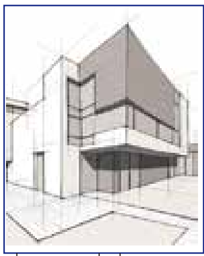
Its the issue that related to Designers in the various field

Q1-Who designed which part?

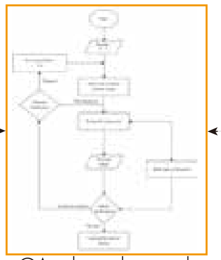
Q2-How much did they do?

Q3-Has the position of a designer changed in such a background?

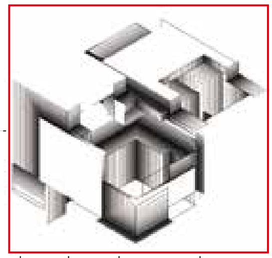
1.2-What I want to test through design



architectural design topic



CA algorithm tech



digital media visualization

my own outcome

Thats quite a topic to be discussed,i have the oppotunity to think&research&give my own answer.

1.3-Further research

searching for some theories to form my own answer to these 3 questions

Q1&Q2

Tools are not neutral.They structure what can be designed before design begins. 2

Machines do not only produce forms.They change the conditions under which forms are imagined. 3

Architectural tools do not simply execute decisions.They structure what can be imagined, represented and decided. 4

A1&A2:
Algorithm tech&AI are like other tools to provide a new platform to explore humans creation&production. While human should imagine and explore new possibilities based on new platform.

Q3

The architect no longer only produces a fixed form, but defines a system of rules through which forms can emerge. 5

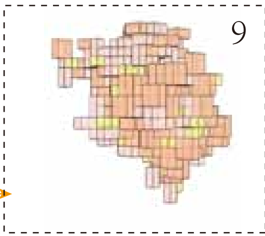
Creativity, likewise, is less a solitary act of invention than a negotiation within systems of shared intelligence. 6

If the algorithm's creator occupies the core position of creativity, then sexuality has not disappeared but has been reorganized. 7

A3:Designers(eg,architects) still play a role as a negotiator that negotiate with algorithm tech by defining new algorithmic rules,new algorithmic systems, new algorithmic shapes,functions,etc ,and take the responsibility of these parts during such an age of collective practice.

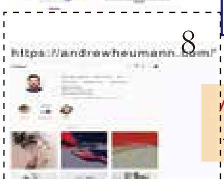
Conclusion:
Starting from the question of the collaborative division of labor between designers and algorithms in the design process of collective practice era
this research inspired me to frame the project as a collective practice that can simulating the negotiation process between contemporary designers and algorithmic technologies,in order to discuss this question.

1.4-Decide the form of this collective practice project

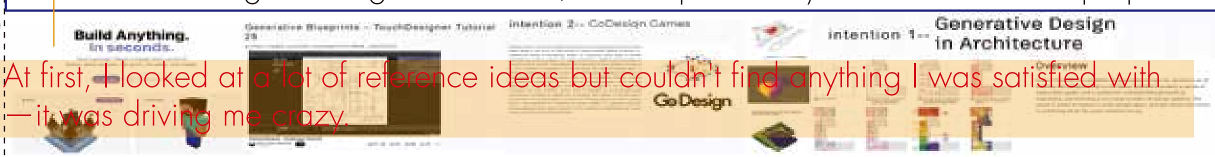


9

After long search, I found a Grasshopper component set that allows the user to define the growth logic themselves, which sparked my initial sketch of the project.

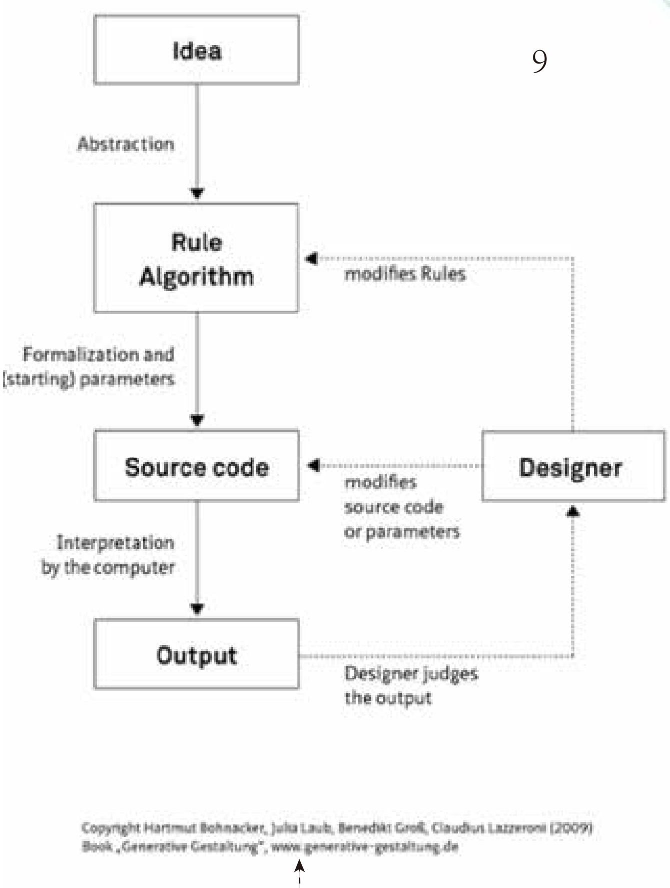


8

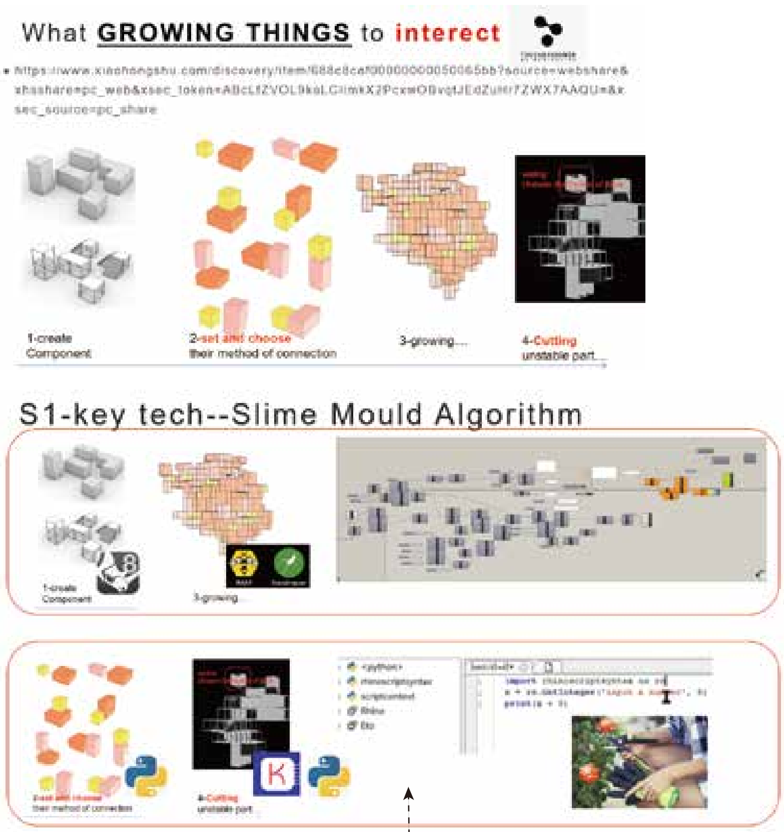


At first, I looked at a lot of reference ideas but couldn't find anything I was satisfied with —it was driving me crazy.

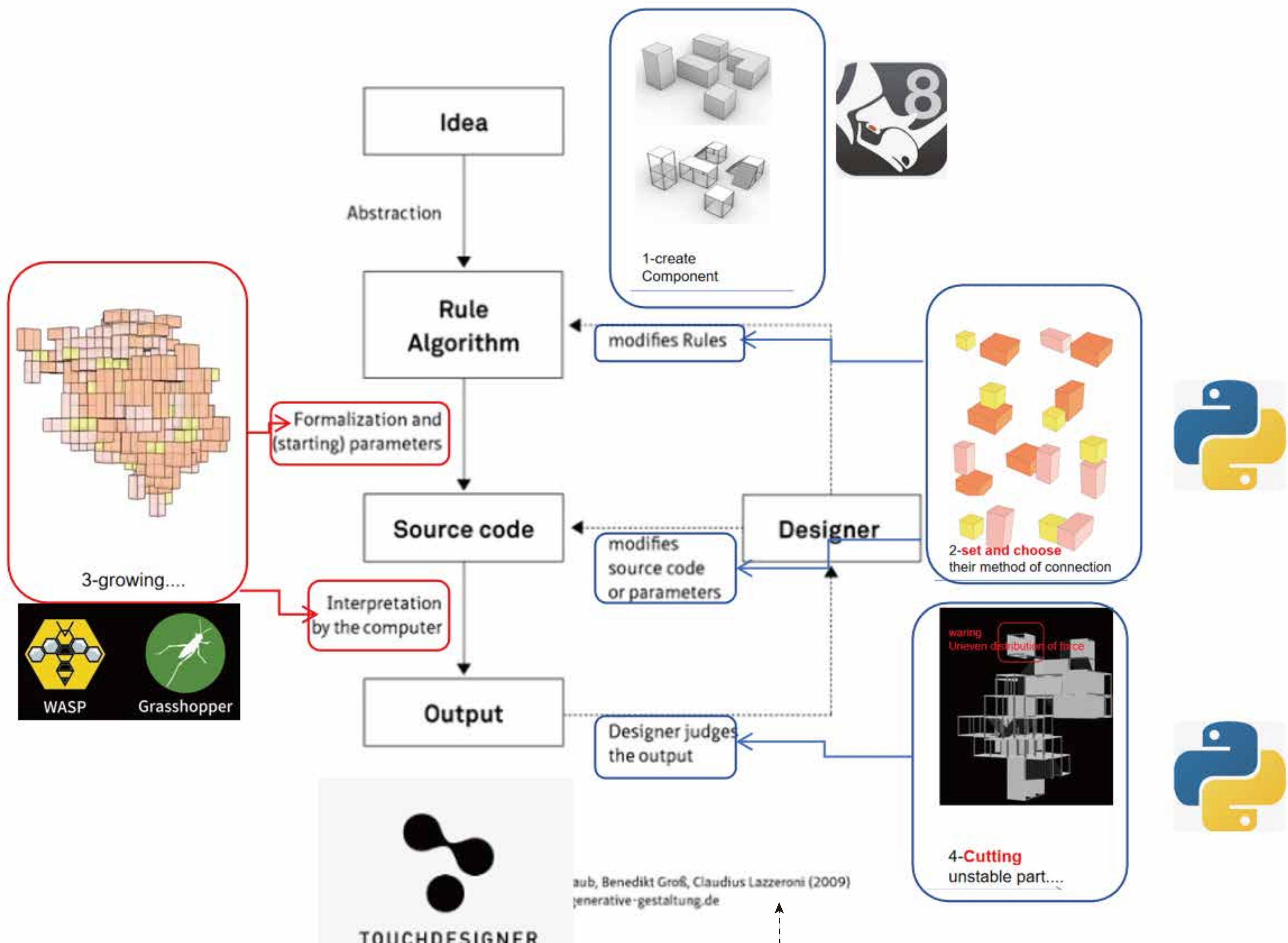
1.5-Deciding the Interection elements & Technical frame



1.51-exploring the process by which contemporary designers negotiate with algorithms to create designs, in order to refine my own interaction process.



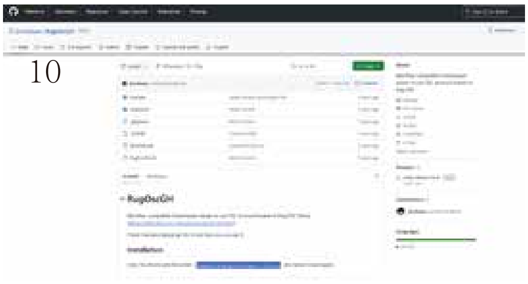
1.52-Seeking an architectural growth algorithm to serve as the primary growth logic.(Rhino GrassHopper)

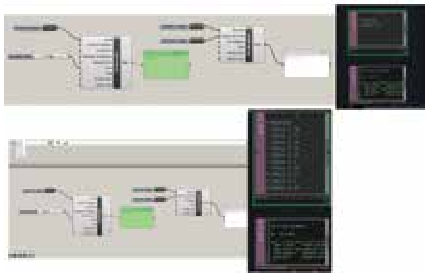


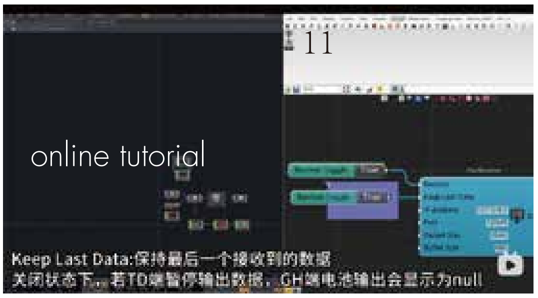
1.52-interection frame base on reference

Technique Experiments

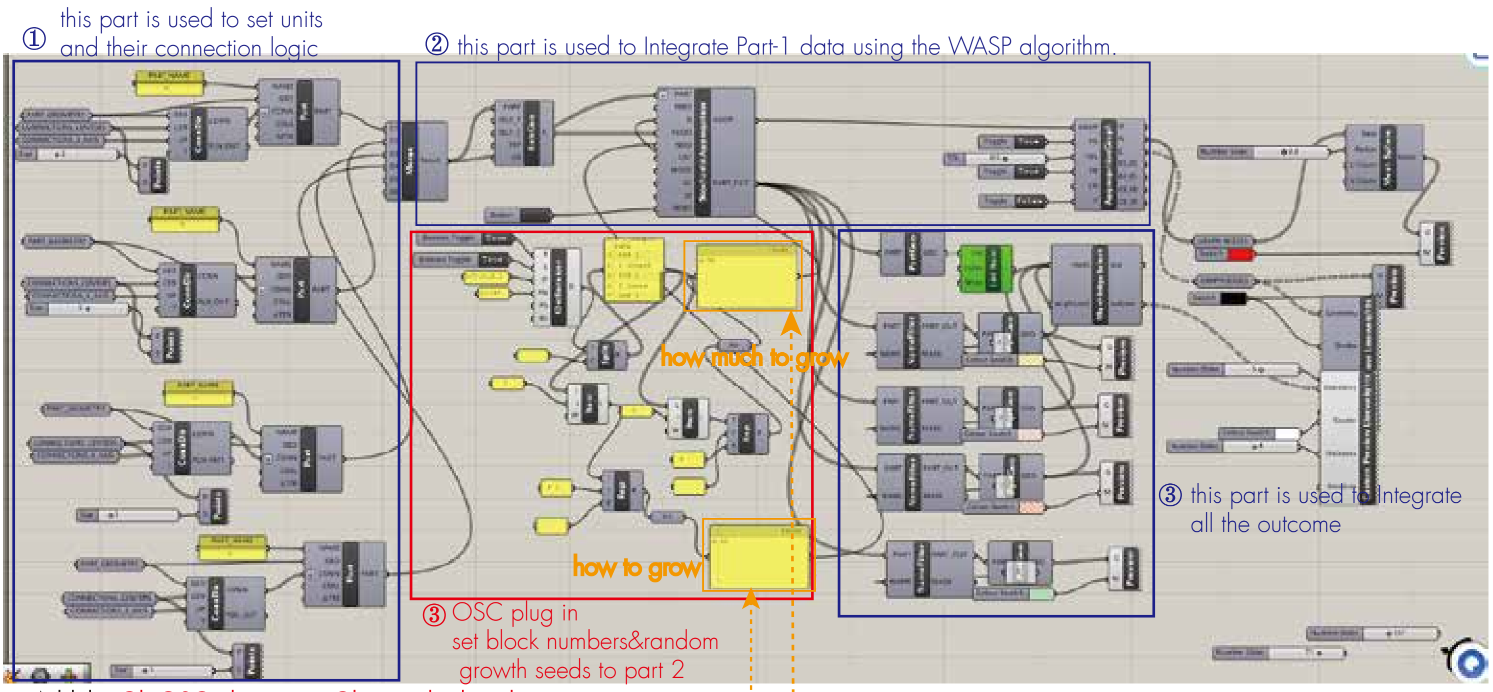
2.1-Rhino Grasshopper-WASP+TD constant&OSC DATA



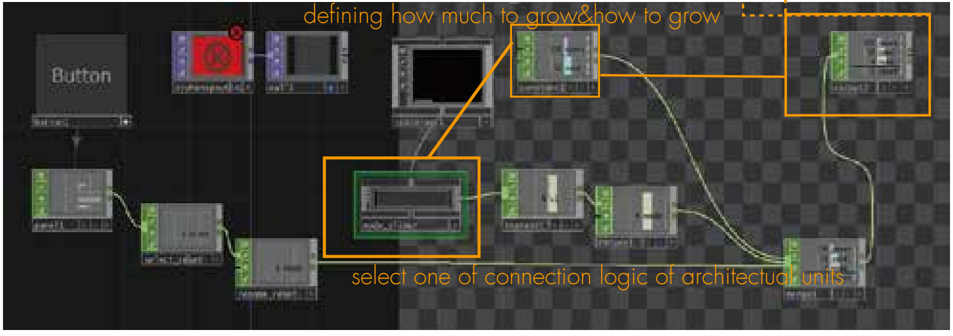


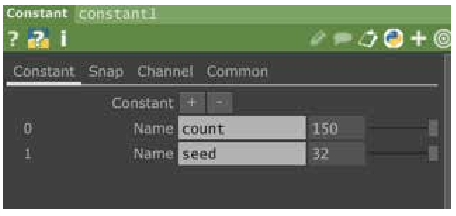


I found a GrassHopper Plug-in that can directly recieve data from TD's OSC DAT & Constant Chop



Add the Gh OSC plug-in into Gh growth algorithm
to be my basic frame of GH part.





Try visualizing the components that control the building's growth within TD.

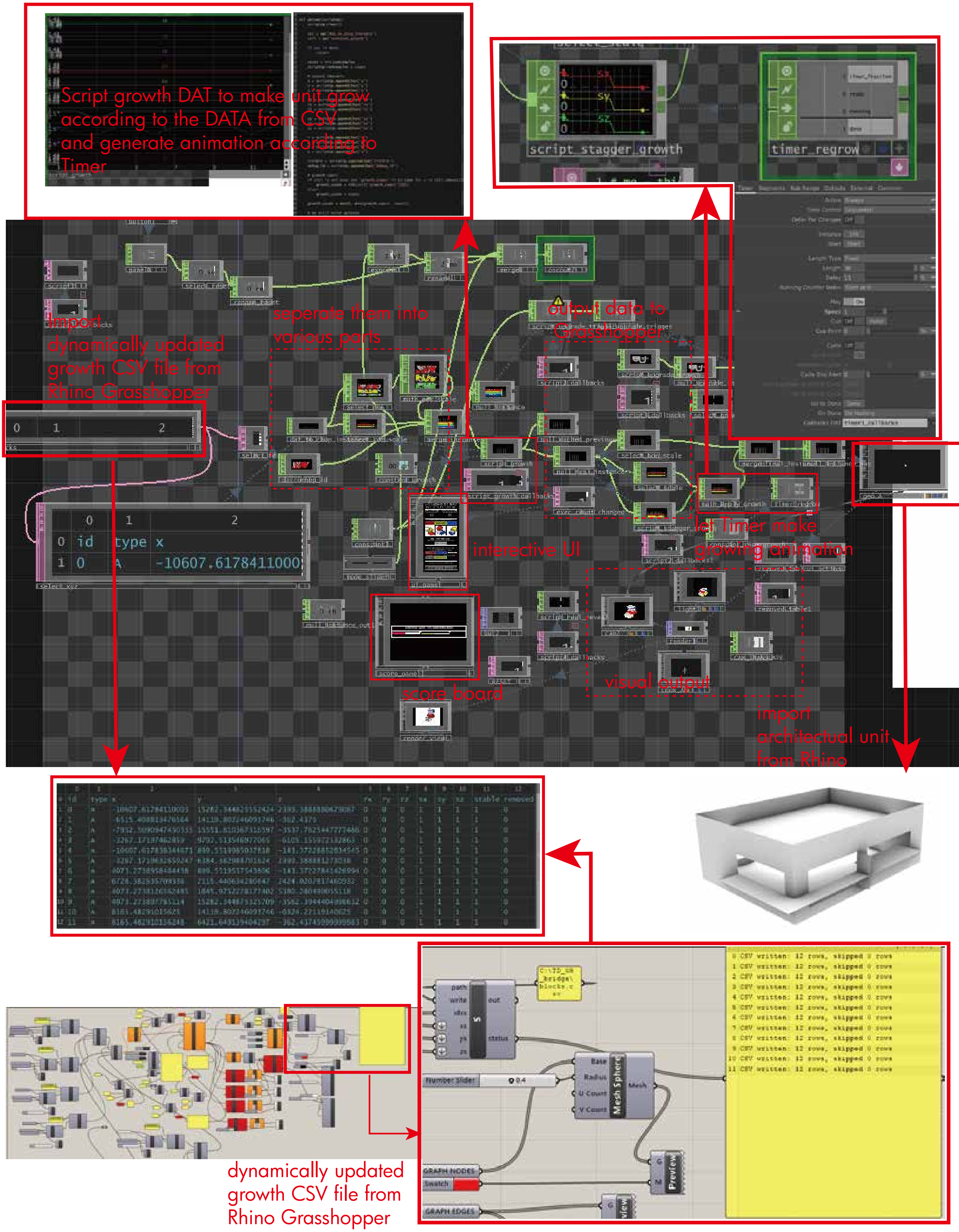
Conclusion:
Now i ve proved that using TD to control the growth of the virtual blocks in GH is accesible.

But so far the system lack the growing animation& triming interection&score board and OBS is not suitable for showing blocks.

I need to make more great modifications to Enrich its visual and interactive experiences.

Final version of interective workflow

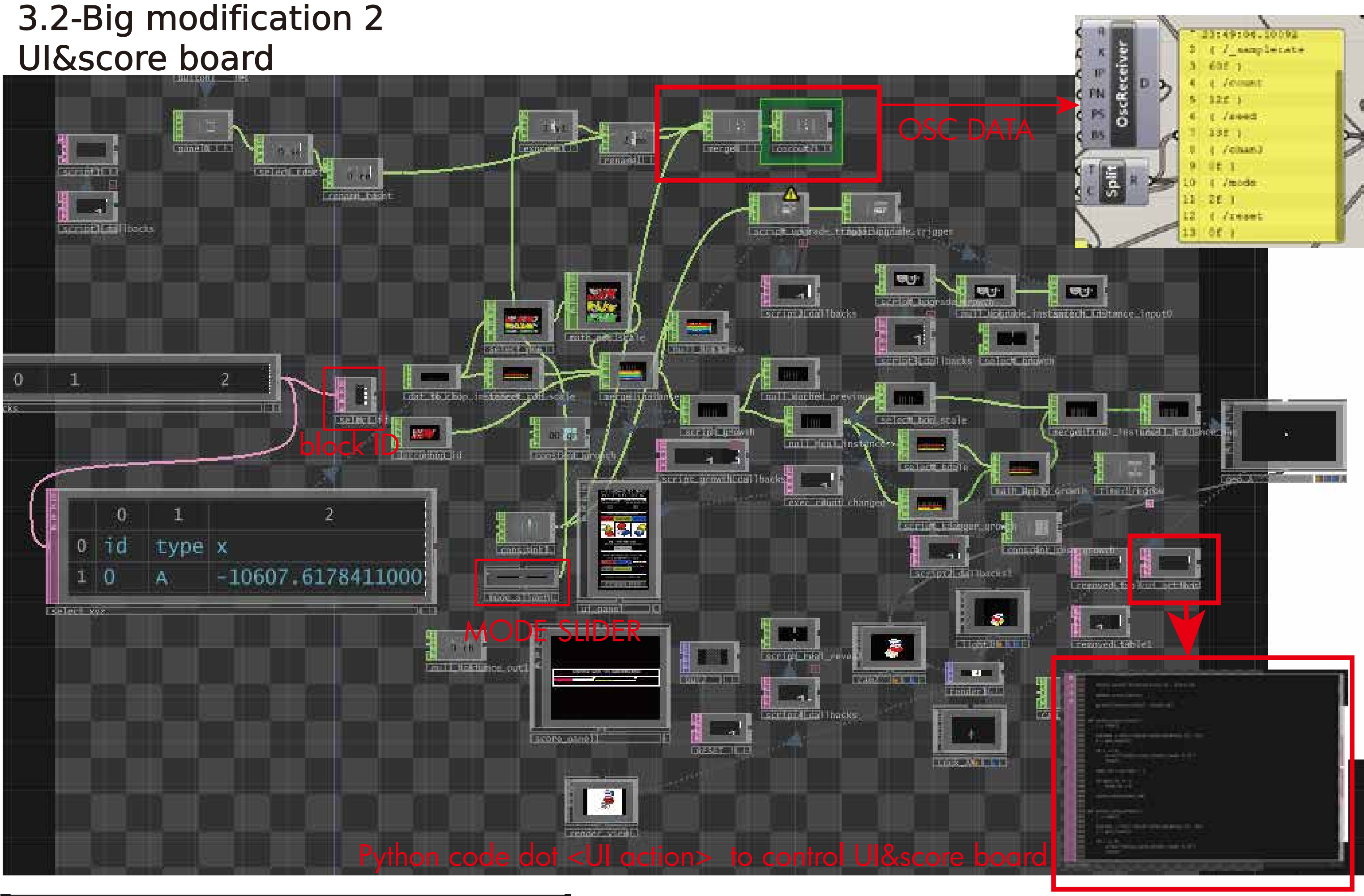
3.1-Big modification 1
setting TD based visualization&growing animation

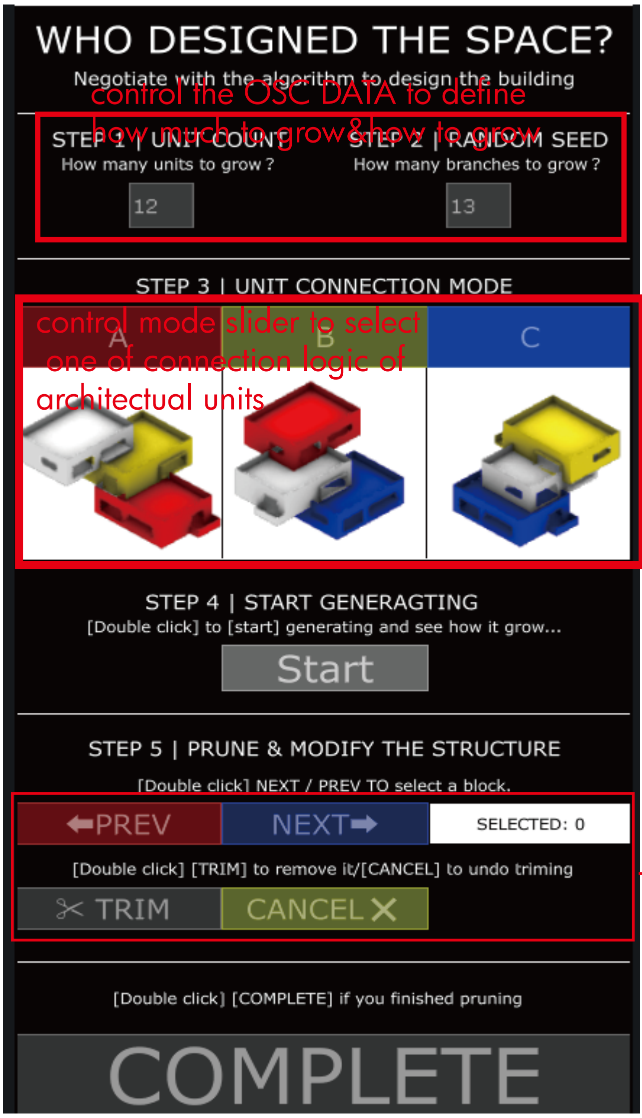


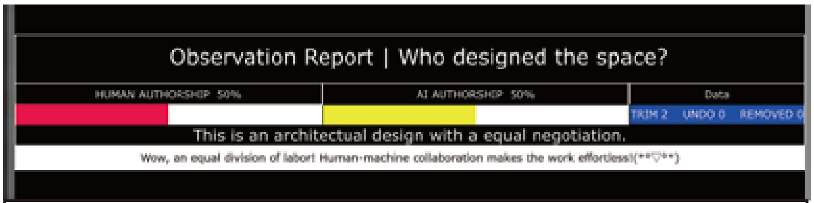
Final version of interective workflow

3.2-Big modification 2

UI&score board

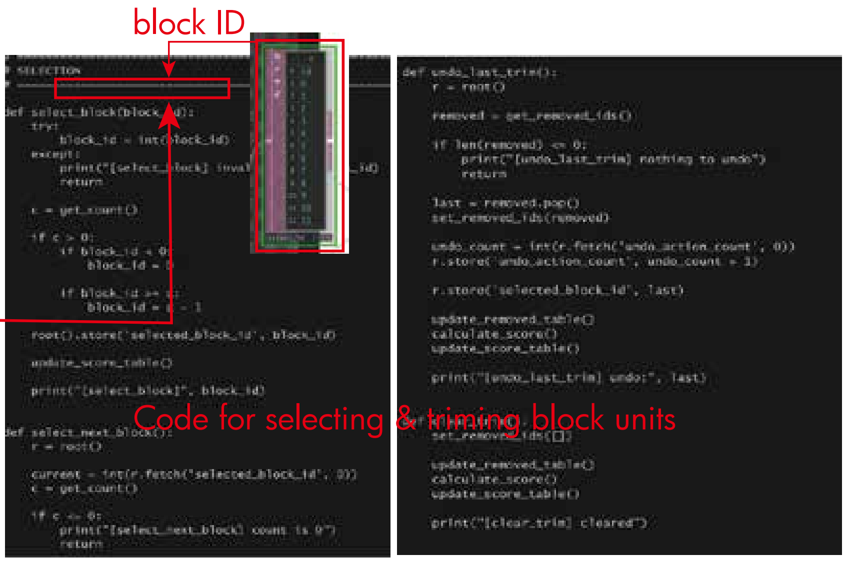




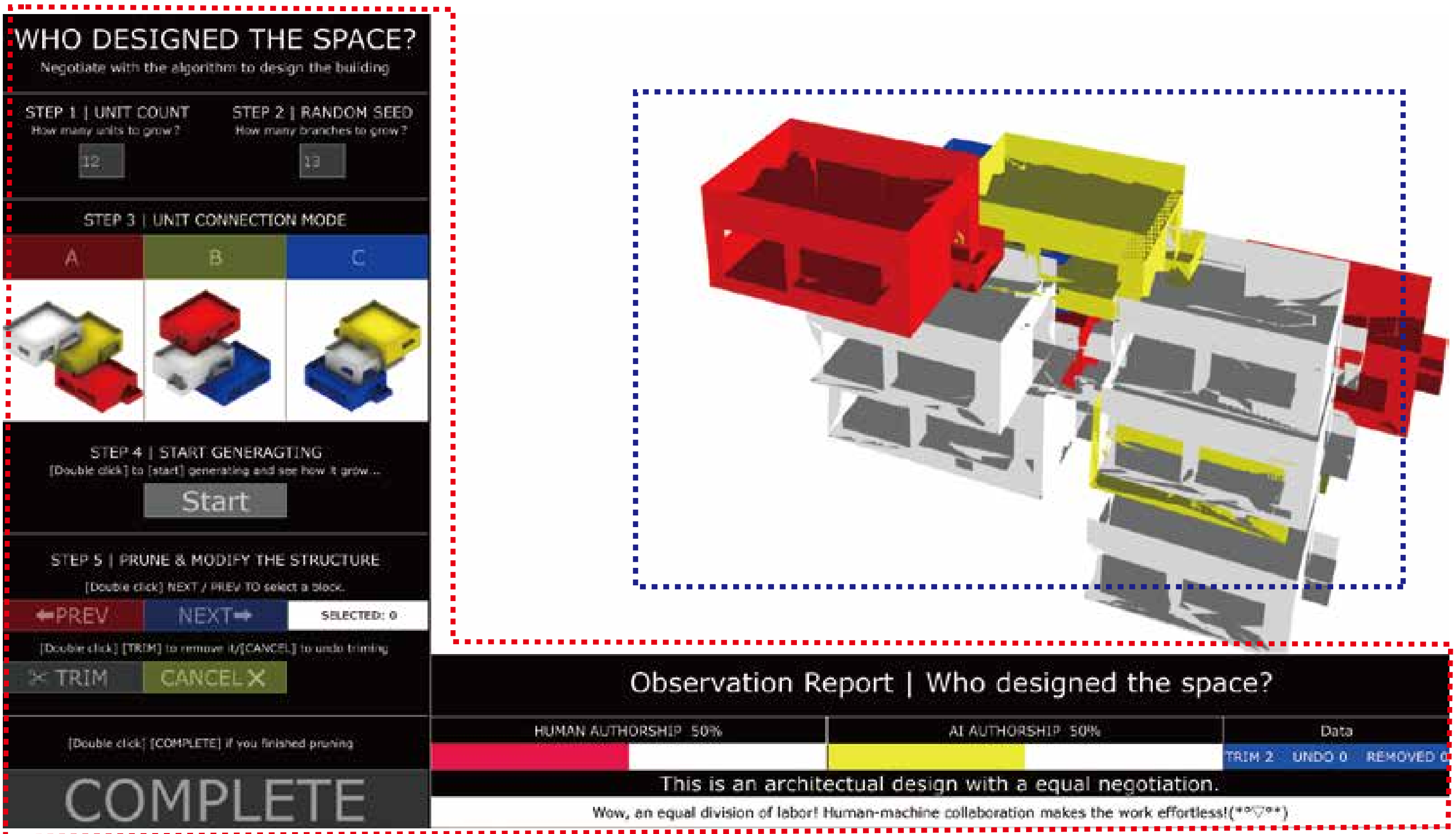


scoring rules and concluding remarks

Scoring is primarily based on the proportion of design authorship by the player versus those determined by the algorithm.
Performing actions STP1 through STP3 awards 10 points per action (considered basic player inputs).
Performing action STP5 awards 15 points per action (considered an advanced maneuver, hence the higher score).
However, since the WASP algorithm is integral to the game's design, the player's maximum possible score is 70 points; a full mark is unattainable.
This demonstrates that even when aided by algorithms, designers must still bear the corresponding responsibilities in their work.



Visual Effects Design



Interface Reference

architectual unit Reference

11 Composition VII (The Three Graces)
Theo van Doesburg

12 Maison Particulière (Private Residence)
Theo van Doesburg&Cornelis van Eesteren

Why did I choose De Stijl(neoplasticism)-style graphic and architectural design as the primary visual language?

1-With a minimalist form, blocky, cubic geometry, and a rich color palette, it conveys a strong sense of rational design.

I believe this rational aesthetic aligns perfectly with my design theme:
1- First, it embodies a modernist architectural style that creates a compelling visual effect of "growth" when combined with Grasshopper (reminiscent of LEGO).
2- Many works of Neoplasticism emerged during the Second Industrial Revolution, representing an avant-garde design movement driven by the era's emerging technologies (machines). This parallels the current trend where algorithmic design is spearheading a new design movement—characterized by a blend of rationality and rules that nonetheless allows for rich, dynamic variation.